VENDRA - Production Deployment Changes

Backend: Render (https://vendra-kxjb.onrender.com) | Frontend: Vercel (https://vendra-woad.vercel.app) | Google login supported.

1. Summary: why changes were needed

The project worked locally because Vite proxied every **/api** request to localhost:3000, so the browser saw a single origin. In production the frontend and backend live on two different domains, and several things were hard-coded for localhost. Without changes: API calls on Vercel would 404, Google login would redirect to localhost, cookies would not be secure, and refreshing a React route would 404.

2. Architecture used: Vercel proxies /api to Render

Frontend/vercel.json contains a rewrite that forwards every **/api/*** request to the Render service. The browser only talks to vendra-woad.vercel.app, so:

• Cookies are first-party (not blocked as third-party cookies by Chrome/Safari).

• No CORS problems for normal API calls.

• Frontend code did not need to change: axios still uses /api/... and the Google button still goes to /api/auth/google.

• The Google redirect URI is the **Vercel** URL, so the login cookie is set on the Vercel domain.

3. Changes, file by file

File	Change	Reason
Frontend/vercel.json (new)	Rewrite /api/(.*) to https://vendra-kxjb.onrender.com/api/$1; every other path to /index.html	Replaces the Vite dev proxy. SPA fallback fixes 404 when refreshing pages like /cart.
Backend/src/config/config.js	Added FRONTEND_URL, GOOGLE_CALLBACK_URL (defaults: localhost / relative path) and REDIS_TLS. In production, FRONTEND_URL and GOOGLE_CALLBACK_URL are required	Removes hard-coded localhost; server fails fast if misconfigured
Backend/src/app.js	CORS enabled (origin FRONTEND_URL, credentials); trust proxy; GET /health; Google strategy callbackURL from env with proxy:true	Safe behind Render's proxy; Render health check; correct Google redirect URI
Backend/src/controllers/user.controller.js	Cookie options (httpOnly, secure in production, sameSite lax, 7 days); Google redirect to FRONTEND_URL; clearCookie with same options; missing return added in register/login; duplicate user returns 409; Google callback fixed (see section 4)	Secure cookies over HTTPS; logout really clears the cookie; prevents crashes; Google login works
Backend/src/routes/auth.route.js	Google failureRedirect = FRONTEND_URL/login	Was localhost or a relative path
Backend/src/dao/cache.js	Optional TLS for Redis when REDIS_TLS=true	Cloud Redis providers usually need TLS
Backend/package.json	engines: node >= 20 (start script already existed)	Render runs npm start
Backend/.env.example (new)	Template listing all environment variables	Reference only; the app never reads it

4. Problems found during deployment and how they were fixed

4.1 Products not loading on the Vercel site

Symptom: 'The store is getting ready', network tab showed products/ with status 404.
Cause: the frontend calls /api/products/ (with a trailing slash). The first rewrite /api/:path* did not match the trailing slash, so Vercel's catch-all returned index.html instead of JSON.
Fix: the rewrite source was changed to /api/(.*) with destination .../api/$1.
Note: one screenshot was taken on a Vercel preview URL (vendra-gupwzioxu-...vercel.app). Preview URLs are locked by Vercel Deployment Protection and are not configured for Google/CORS. Always use vendra-woad.vercel.app.

4.2 Google error: redirect_uri_mismatch

Cause: Google Cloud Console did not yet contain the production callback.
Fix (Google Cloud Console, no code change): Authorized redirect URI = https://vendra-woad.vercel.app/api/auth/google/callback and Authorized JavaScript origin = https://vendra-woad.vercel.app (localhost entries kept).

4.3 Google login returned 'Internal Server Error'

Render log: MongoServerError E11000 duplicate key, index fullname_1, value 'Rishi Maheshwari'.
Cause: the user model marks fullname as unique. An account with that name already existed under a different email, so creating the Google user failed and Express returned 500. (The later TokenError invalid_grant in the log was just the browser re-using an already-used Google code and can be ignored.)
Fix in googleCallback: if the name is taken, a short suffix (last 4 characters of the Google id) is added to the name; the whole callback is wrapped in try/catch so any error is logged and redirects to FRONTEND_URL/login instead of showing a 500; an unused photos[0].value line that could crash was removed.

5. Environment variables

Render (Environment):

```
NODE_ENV=production
FRONTEND_URL=https://vendra-woad.vercel.app
GOOGLE_CALLBACK_URL=https://vendra-woad.vercel.app/api/auth/google/callback
MONGO_URI, JWT_SECRET, GOOGLE_CLIENT_ID, GOOGLE_CLIENT_SECRET,
IMAGEKIT_PRIVATE_KEY, RAZORPAY_KEY_ID, RAZORPAY_KEY_SECRET,
REDIS_HOST, REDIS_PORT, REDIS_PASSWORD   (+ REDIS_TLS=true if required)
Do NOT set PORT (Render provides it).
```

Vercel (Environment): VITE_RAZORPAY_KEY_ID (must match the Razorpay key used on Render). No API base-URL variable is needed.

Local .env: unchanged. GOOGLE_CALLBACK_URL=http://localhost:5173/api/auth/google/callback still works for local development; it is never sent to Render.

6. Platform settings

Platform	Setting
Render	Root Directory: Backend | Build: npm install | Start: npm start | Health Check Path: /health
Vercel	Root Directory: Frontend | Framework: Vite | Build: npm run build | Output: dist
MongoDB Atlas	Network Access must allow Render (0.0.0.0/0 on the free plan)
Google Cloud Console	Redirect URI and JavaScript origin for https://vendra-woad.vercel.app (see 4.2)

7. How to verify production

• https://vendra-kxjb.onrender.com/health returns {status: ok}.

• https://vendra-woad.vercel.app/api/products/ returns JSON (served from Render).

• On https://vendra-woad.vercel.app: register, login, logout, Google login, add to cart, Razorpay payment, create product with image upload.

8. Things to remember

• Render's free plan sleeps when idle; the first request can take about a minute.

• If you change the Render URL, update Frontend/vercel.json. If you change the Vercel domain, update FRONTEND_URL, GOOGLE_CALLBACK_URL and the Google Console entries.

• Never commit .env or share secret values.

• Local development is unchanged: new settings default to localhost and the Vite proxy still works.

• Google users whose name already exists get a suffix. Old accounts are not merged; Google login matches accounts by email.